# Benchmarking LLMs on Advertising Slogan Writing

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## 1 Implementation and reproducibility

We graded five LLMs on writing one slogan from a company brief. The eval set is **50 real companies**: 28 standard briefs, 12 with word-limit and required/banned-word constraints, and 10 edge cases (sparse description, sensitive industry, long irrelevant context, injected instruction). Twelve items are dev; 38 are held-out and carry the headline numbers. On 2026-10-12 we ran claude-sonnet-5.5, gemini-3.8-flash, gpt-6.1-sol, mistral-large-4-0 and deepseek-v4.1-flash through OpenRouter with an identical prompt, fresh context and **5 runs per item** (1,325 outputs); no rubric or examples were supplied. Outputs get rule checks plus a blinded judge (grok-4.7, never a generator); claude-opus-5.5 re-scores held-out outputs to test self-preference. **Deviations from the proposal:** slogans not jokes; one prompt condition, not three; 50 items × 5 runs, not 60 × 20; a new model and judge set; an added unsupported-claim criterion and rule checks; a held-out split; a 60-output human audit; Mistral replacing the age-gated Muse Spark. Temperature could not be held fixed: Sonnet and GPT accept none. The archive holds the eval set, judge prompt, grading code and raw outputs, and re-runs from its README.

## 2 Results

| Model | Valid, % | Originality (1–5) | Constraints met, % | Median latency | Gen. cost |
|---|---|---|---|---|---|
| Claude Sonnet 5.5 | 85.8 ± 8.4 | 2.65 ± 0.08 | 78 ± 13.0 | 1.5 s | $0.16 |
| Gemini 3.8 Flash | 96.8 ± 3.4 | 2.58 ± 0.13 | 98 ± 4.5 | 2.0 s | $0.18 |
| GPT-6.1 Sol | 98.9 ± 1.4 | 2.54 ± 0.09 | 100 ± 0.0 | 3.2 s | $0.25 |
| Mistral Large 4 | 97.9 ± 1.2 | 2.37 ± 0.09 | 100 ± 0.0 | 20.3 s | $0.86 |
| DeepSeek V4.1 Flash | 98.4 ± 2.4 | 2.56 ± 0.07 | 100 ± 0.0 | 2.7 s | $0.09 |

Held-out items, mean ± SD across 5 runs; line: valid ≥ 95%, originality ≥ 3.0, constraints ≥ 90%. Gemini, GPT, Mistral and DeepSeek are **not separable** on validity or constraints (gaps inside run-to-run noise), so we do not rank them; only Sonnet separates, lower. Edge cases: Sonnet falls to 76% on constraint briefs and 71% on stress items; the others stay at 94–100%. Mistral is 14× slower than the fastest model because of ~1,500 hidden reasoning tokens per slogan. Total generation cost was $1.54; grading cost $13.23.

## 3 Analysis and interpretation

**No model clears the line**: all five miss originality (2.37–2.65 vs 3.0) and Sonnet also misses validity (85.8%) and constraints (78%). **Model wrong:** 25 of 190 Sonnet outputs broke a word limit stated in the prompt (other models: 1 or fewer). **Missing input:** on the three sparse items, supplying the full description cut generic outputs from 53–73% to 0–27% and lifted originality 0.6–1.1 points, so much of that shortfall is the brief, not the model. **Instrument:** contamination bit: Mistral reproduced a real company slogan in 20% of outputs and the judge scored a verbatim Coke slogan 4/5, so originality is inflated by memorization (we report a supplementary score without it). Yes/no criteria saturate (≥ 97% yes), so validity differences come from our rules, not the judge, and the two judges agree only moderately on originality (QWK 0.64). We found **no self-preference** (Claude judge minus neutral judge -0.12, 95% CI -0.24 to +0.00). The blinded human audit is prepared but not yet scored, so judge validity is untested against people.

**Follow-up, verbalized sampling (exploratory; 12 held-out items, 5 candidates each, pre-registered afterwards).** Asking for five slogans in one call raised distinct ideas from 1.9–3.1 (five independent calls) to 3.2–3.6 for a plain list and 3.3–3.8 for verbalized sampling (of 5). The probability request added only -0.25 to +0.50 over the list (every 95% interval includes zero) and did not raise originality (-0.14 to -0.06), so the pre-registered rule marks it beneficial for none of the five models. The gain comes from list-style prompting, not from the probabilities.

## 4 Limitations and what we would do next

This benchmark does not show that a slogan sells: originality is a model's rating, not audience response; 50 companies and one prompt cannot cover brand voice; the stress subtypes have 2–3 items each; sampling parameters differed by model; and famous companies make memorization likely. With two more weeks we would make one change: replace the LLM originality score with blinded human ratings of about 200 slogans and test whether either score predicts which slogans people choose.